C & C++ Question Paper

Date : 8-02-2024
Marks : 50

Write the Output for the following C programs.

1)
```
#include<stdio.h>
int main()
{
    char ch = 350;
    printf("ch = %d\n",ch);
}
```
94

2)
```
#include<stdio.h>
int main()
{
    int x = 48;
    int y = 0x48;
    int z = 048;
    printf("%d %d %d\n",x,y,z);
}
```
→ error

3)
```
#include<stdio.h>
int main()
{
    printf("%ld %ld %ld\n",sizeof(4),sizeof('4'),sizeof("4"));
}
```
4 4 2

4)
```
#include<stdio.h>
int main()
{
    printf("%d\n",printf("%f",3.0));
}
```
3.000000 8

5)
```
#include<stdio.h>
int main()
{
    float f1 = 22.7, f2 = 3.14;
    f1 = f1^f2;
    f2 = f1^f2;
    f1 = f1^f2;
    printf("f1 = %f  f2 = %f\n",f1,f2);
}
```
error

6)
```
#include<stdio.h>
int main()
{
    extern int x;
    printf("x = %d\n",x);
}
static int x = 10;
```

error

7)
```
#include<stdio.h>
int main()
{
    char *ptr = "Compassion";
    char arr[] = "Compassion";

    if(ptr == arr)
    printf("Equal\n");
    else
    printf("Not Equal\n");
}
```

Not equal

8)
```
#include<stdio.h>
int main()
{
    if(!printf("Hello...\n"))
    printf("Hai...\n");
    else {
    printf("Bye...\n");
    break;
    }
}
```

error

9)
```
#include<stdio.h>
int main()
{
    int i,a[] = {10,20,30,40,50};
    char *p = (char *)a;

    for(i=0;i<5;i++)
    printf("%d ",p[i]);
    printf("\n");
}
```

10 0 0 0 20

10)
```
#include<stdio.h>
int main()
{
    int x = 300;
    char *p = (char *)&x;
    int **ptr = (int **)&p;
```

```
    printf("x = %d  *p = %d  **ptr = %d\n",x,*p,**ptr);
}
```

300 44 44

11)
```
#include<stdio.h>
#include<string.h>
int main()
{
    char str[] = "abcdef";
    char *ptr = str+strlen(str)-1;

    printf("%c\n",*ptr--);
    printf("%c\n",*--ptr);
    printf("%c\n",(*ptr)--);
    printf("%c\n",--*ptr);
    printf("%c\n",--*--ptr);
    printf("%s\n",str);
}
```

f
d
b
b
b
abbbef

12)
```
#include<stdio.h>
#include<string.h>
int main()
{
    char s1[20] = "Honesty", s2[20] = "Loyalty";
    char str[20] = "Onward";

    char *ptr = strcat(str,strcat(s2,s1));
    printf("%s\n",ptr);
}
```

OnwardLoyaltyHonesty

13)
```
#include<stdio.h>
int main()
{
    char *p[] = {"Promise","Tolerance","Character","Responsibility"};
    char **ptr = p;
    ptr += 2;
    printf("%s\n",(*ptr)+4);
}
```

acter

14)
```
#include<stdio.h>
struct st
{
    char a:3;
    char b:3;
    char c:3;
    char d:3;
    char e:3;
};
```

```
int main()
{
   printf("size : %ld\n",sizeof(struct st));  3
}
```

15)
```
#include<stdio.h>
union U
{
    struct st1
    {
        char c[2];
        char ch[2];
    }v1;
    struct st2
    {
        short int x;
        short int y;
    }v2;
}var = {12,1,15,1};
int main()
{
    printf("%d  %d\n",var.v2.x,var.v2.y);
}
```

268 271

16) What is the sizeof long int on 32-bit machine ? (1M) 4 bytes
17) What is FILE in C program ? (1M)
18) What is Memory leak in DMA? How to avoid it ? (1M)
19) What is Recursion function? What is the merit and demerit of it? (2M)
20) Write a generic program to swap 2 variables data (for all data types) (3M)
21) WAP to check the number of bits are set and clear in a given integer number. (3M)
22) Design an user defined function for strncpy( ) (4M)
23) Write a program to implement **cp** command. (5M)
24) What are the access specifiers in C++? Write one line statement for each one. (3M)
25) Write the syntax for pre & post-increment operators using friend function. (2M)
26) What is Virtual function? Explain with one example. (5M)
27) What are the default member functions of a class provided by compiler? (2M)
28) Write the differences between new & malloc. (3M)

Module: ARM Total Marks : 15

A) MCQ's

1) According to upper limit frequency in LPC2129 the multiplier value cannot exceed _____.

a) 4 b) 32 c) 6 d) 5

2) An _____ is any condition that needs to halt the normal sequential execution of instructions.

a) Duty cycle b) Exception c) Baudrate d) None of the above

3) The 32 bit PC is a counter which is incremented to the value stored in _____.

a) T0TCR b) PR c) TC d) TCR

4) Every non-vectored interrupts have _____ address.

a) Unique b) Common c) Both 1 & 2 d) None of the above

5) How many Display control pins are there in Lcd.

a) 3 b) 2 c) 8 d) 16

6) _____ is the slect starting address of CG RAM in LCD.

a) 0x80 b) 0x40 c) 0x28 d) 0x0C

7) The standard data frame format in UART is _____frame.

a) 4N1 b) 8N1 c) 16N2 d) 8N2

8) CAN protocol was developed by _____.

a) Motorola b) Philips(NXP) c) INTEL d) Robert Bosch

9) In UART data transfer or recieve always _____bit first.

a) LSB b) MSB c) LSB OR MSB d) None of the above

10) Standard speed In I2C protocol is_____ .

a) 1 Mbps b) 5 Mbps c) 3.4 Mbps d) 100Kbps

11) SPI protocol was developed by _____.

a) Motorola b) Philips(NXP) c) INTEL d) SAMSUNG

12) LPC 2129 is a _____ bit Microcontroller.

a) 32 b) 24 c) 8 d) 64

13) In high-level sensitivity as long as the EINT0 pin is _____ ,the EINT0 flag not allowed to clear.

a) high b) low c) None of the above

14) _____ protocol is the example for asynchronous serial communication.

a) I2C b) SPI c) CAN d) All of the above

15) I2C supports _____mode of communication.

a) Full duplex b) Half duplex c) Simplex d) All of the above

APTITUDE COMPANY SPECIFIC TEST (HBC-0802)

NAME: VECTOR ID:

1. At what angle the hands of a clock are inclined at 15 minutes past 5?
A. 52.5° B. 67.5° C. 60° D. None

2. A trader bought a car at 20% discount on its original price. He sold it at 40% increase on the price he bought it. What percent of profit did he make on the original price?
A. 22% B. 32% C. 12% D. None

3. If the first day of the year(other than the leap year) was Friday, then which was the last day of the year?
A. Monday B. Thursday C. Friday D. None

4. The difference between simple and compound interests compounded annually on a certain sum of money for 2 years at 4% per annum is Re 1. The sum is:
A. Rs 625 B. Rs 630 C. Rs640 D. None

5. A and B can do job in 8 days. B and C can do same job in 12 days. A, B and C together can do same job in 6 days. In how many days A and C together can complete the job?
A. 4 days B. 8 days C. 12 days D. None

6. Two pipes A and B can fill a cistern in 15 hours and 10 hours respectively. A tap C can empty the full cistern in 30 hours. All the three taps were open for 2 hours, when it was remembered that the emptying tap had been left open. It was then closed. How many hours more would it take for the cistern to be filled?
A. 6 hr 30 min. B. 5 hr 12 min C. 4 hr 24 min. D. None

7. Gauri went to the stationers and bought things worth Rs.25, out of which 30 paise went on sales tax on taxable purchases. If the tax rate was 6%, then what was the cost of the tax free items?
A. Rs.15 B. Rs.15.70 C. Rs.19.70 D. None

8. Aruna cut a cake into two halves and cuts one half into smaller pieces of equal size.Each of the small pieces is twenty grams in weight. If she has seven pieces of the cake in all with her, how heavy was the original cake ?
A. 140 grams B. 240 grams C. 280 grams D. None

9. A sum of Rs. 12,000 deposited at compound interest becomes double after 5 years. After 20 years, it will become:
A. Rs. 1,10,000 B. Rs. 1,30,000 C. Rs. 1,24,000 D. None

10. A batsman has a certain average of runs for 16 innings. In the 17th innings, he makes a score of 85 runs, thereby increasing his average by 3. What is the average after the 17th innings?
A. 60 B. 37 C. 38 D. None

Communicaton Question Paper

Each Question :5Mrks

1) Describe about yourself. (5-7 lines)

2) What is the toughest movement you have faced in your life and How did you overcome it using any of your strength (10-12 lines)

3) Why have you choosen the field of Embedded (5-8 lines)

4) Have you ever worked in a team, If yes then, explain the most challenging situation. (5-8 lines)

5) Why should any embedded company recruit you? (5-8 lines)

6) What will you contribute to the company within sixty day's of your hiring? (5-8 lines)